

ANSI SQL Using MySQL Exercises

Database Schema

1. Users

Column Name	Data Type	Constraints	Description
user_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique user identifier
full_name	VARCHAR(100)	NOT NULL	User's full name
email	VARCHAR(100)	UNIQUE, NOT NULL	User's email
city	VARCHAR(100)	NOT NULL	City of the user
registration_date	DATE	NOT NULL	Date when user registered

2. Events

Column Name	Data Type	Constraints	Description
event_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique event identifier
title	VARCHAR(200)	NOT NULL	Title of the event
description	TEXT		Event description
city	VARCHAR(100)	NOT NULL	City where event is held
start_date	DATETIME	NOT NULL	Start date and time of event
end_date	DATETIME	NOT NULL	End date and time of event
status	ENUM	('upcoming','completed','cancelled')	Current status of the event
organizer_id	INT	FOREIGN KEY → Users(user_id)	User who organized the event

3. Sessions

Column Name	Data Type	Constraints	Description
session_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique session identifier
event_id	INT	FOREIGN KEY → Events(event_id)	Associated event
title	VARCHAR(200)	NOT NULL	Title of the session
speaker_name	VARCHAR(100)	NOT NULL	Name of speaker
start_time	DATETIME	NOT NULL	Session start time
end_time	DATETIME	NOT NULL	Session end time

4. Registrations

Column Name	Data Type	Constraints	Description
registration_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique registration ID
user_id	INT	FOREIGN KEY → Users(user_id)	User who registered
event_id	INT	FOREIGN KEY → Events(event_id)	Registered event
registration_date	DATE	NOT NULL	Date of registration

5. Feedback

Column Name	Data Type	Constraints	Description
feedback_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique feedback ID
user_id	INT	FOREIGN KEY → Users(user_id)	User who gave feedback
event_id	INT	FOREIGN KEY → Events(event_id)	Event for which feedback is given
rating	INT	CHECK (rating BETWEEN 1 AND 5)	Rating out of 5
comments	TEXT		Optional comments
feedback_date	DATE	NOT NULL	Date of feedback

6. Resources

Column Name	Data Type	Constraints	Description
resource_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique resource ID
event_id	INT	FOREIGN KEY → Events(event_id)	Associated event
resource_type	ENUM	('pdf','image','link')	Type of resource
resource_url	VARCHAR(255)	NOT NULL	Path or link to resource
uploaded_at	DATETIME	NOT NULL	Upload timestamp

Sample Dataset

1. Users

user_id	full_name	email	city	registration_date
1	Alice Johnson	alice@example.com	New York	2024-12-01

2	Bob Smith	bob@example.com	Los Angeles	2024-12-05
3	Charlie Lee	charlie@example.com	Chicago	2024-12-10
4	Diana King	diana@example.com	New York	2025-01-15
5	Ethan Hunt	ethan@example.com	Los Angeles	2025-02-01

2. Events

event_id	title	description	city	start_date	end_date	status	organizer_id
1	Tech Innovators Meetup	A meetup for tech enthusiasts.	New York	2025-06-10 10:00:00	2025-06-10 16:00:00	upcoming	1
2	AI & ML Conference	Conference on AI and ML advancements.	Chicago	2025-05-15 09:00:00	2025-05-15 17:00:00	completed	3
3	Frontend Development Bootcamp	Hands-on training on frontend tech.	Los Angeles	2025-07-01 10:00:00	2025-07-03 16:00:00	upcoming	2

3. Sessions

session_id	event_id	title	speaker_name	start_time	end_time
1	1	Opening Keynote	Dr. Tech	2025-06-10 10:00:00	2025-06-10 11:00:00
2	1	Future of Web Dev	Alice Johnson	2025-06-10 11:15:00	2025-06-10 12:30:00
3	2	AI in Healthcare	Charlie Lee	2025-05-15 09:30:00	2025-05-15 11:00:00
4	3	Intro to HTML5	Bob Smith	2025-07-01 10:00:00	2025-07-01 12:00:00

4. Registrations

registration_id	user_id	event_id	registration_date
1	1	1	2025-05-01
2	2	1	2025-05-02
3	3	2	2025-04-30
4	4	2	2025-04-28

5	5	3	2025-06-15
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5. Feedback

feedback_id	user_id	event_id	rating	comments	feedback_date
1	3	2	4	Great insights!	2025-05-16
2	4	2	5	Very informative.	2025-05-16
3	2	1	3	Could be better.	2025-06-11

6. Resources

resource_id	event_id	resource_type	resource_url	uploaded_at
1	1	pdf	https://portal.com/resources/tech_meetup_agenda.pdf	2025-05-01 10:00:00
2	2	image	https://portal.com/resources/ai_poster.jpg	2025-04-20 09:00:00
3	3	link	https://portal.com/resources/html5_docs	2025-06-25 15:00:00

Exercises

1. User Upcoming Events

Show a list of all upcoming events a user is registered for in their city, sorted by date.

Query:

```
select u.user_id,u.full_name,e.title,e.start_date
from users u join registrations r on u.user_id=r.user_id
join events e on r.event_id=e.event_id
where e.status='upcoming'
and e.city=u.city
order by e.start_date;
```

2. Top Rated Events

Identify events with the highest average rating, considering only those that have received at least 10 feedback submissions.

Query:

```
select event_id,avg(rating) as ar from feedback
group by event_id
having count(feedback_id)>=10
order by ar desc;
```

3. Inactive Users

Retrieve users who have not registered for any events in the last 90 days.

Query:

```
select u.user_id,u.full_name from users u
where u.user_id not in
(select distinct user_id
from registrations
where registration_date>=curdate()-interval 90 day );
```

4. Peak Session Hours

Count how many sessions are scheduled between 10 AM to 12 PM for each event.

Query:

```
select event_id,count(session_id) as sc from sessions
where hour(start_time)>=10
and hour(end_time)<=12
group by event_id;
```

5. Most Active Cities

List the top 5 cities with the highest number of distinct user registrations.

Query:

```
select u.city,count(distinct r.user_id) as cnt from users u
join registrations r on u.user_id=r.user_id
group by u.city
order by cnt desc
limit 5;
```

6. Event Resource Summary

Generate a report showing the number of resources (PDFs, images, links) uploaded for each event.

Query:

```
select event_id,  
sum(case when resource_type='pdf' then 1 else 0 end) as pdf_count,  
sum(case when resource_type='image' then 1 else 0 end) as image_count,  
sum(case when resource_type='link' then 1 else 0 end) as link_count from resources  
group by event_id;
```

7. **Low Feedback Alerts**

List all users who gave feedback with a rating less than 3, along with their comments and associated event names.

Query:

```
select u.full_name, f.comments, e.title from users u  
join feedback f on u.user_id=f.user_id  
join events e on e.event_id=f.event_id  
where f.rating<3;
```

8. **Sessions per Upcoming Event**

Display all upcoming events with the count of sessions scheduled for them.

Query:

```
select e.event_id,e.title,count(s.session_id) as session_count from events e  
left join sessions s on e.event_id=s.event_id  
where e.status='upcoming'  
group by e.event_id,e.title;
```

9. **Organizer Event Summary**

For each event organizer, show the number of events created and their current status (upcoming, completed, cancelled).

Query:

```
select u.user_id, u.full_name,  
sum(case when e.status='upcoming' then 1 else 0 end) as upcoming_cnt,  
sum(case when e.status='completed' then 1 else 0 end) as completed_cnt,  
sum(case when e.status='cancelled' then 1 else 0 end) as cancelled_cnt from users u  
join events e on u.user_id=e.organizer_id  
group by u.user_id,u.full_name;
```

10. Feedback Gap

Identify events that had registrations but received no feedback at all.

Query:

```
select event_id from registrations
group by event_id
having count(registration_id)>0
and event_id not in
(select event_id from feedback );
```

11. Daily New User Count

Find the number of users who registered each day in the last 7 days.

Query:

```
select registration_date, count(user_id) as new_users from users
where registration_date>=curdate()-interval 7 day
group by registration_date
order by registration_date;
```

12. Event with Maximum Sessions

List the event(s) with the highest number of sessions.

Query:

```
select e.event_id, count(s.session_id) as sc from events e
join sessions s on e.event_id=s.event_id
group by e.event_id
order by sc desc
limit 1;
```

13. Average Rating per City

Calculate the average feedback rating of events conducted in each city.

Query:

```
select e.city, avg(f.rating) as avg_rating from events e
join feedback f on e.event_id=f.event_id
group by e.city;
```

14. Most Registered Events

List top 3 events based on the total number of user registrations.

Query:

```
select event_id, count(user_id) as registrations from registrations
group by event_id
```

```
order by registrations desc  
limit 3;
```

15. Event Session Time Conflict

Identify overlapping sessions within the same event (i.e., session start and end times that conflict).

Query:

```
select s1.event_id, s1.session_id, s2.session_id from sessions s1  
join sessions s2  
on s1.event_id=s2.event_id  
and s1.session_id<s2.session_id  
and s1.start_time<s2.end_time  
and s1.end_time>s2.start_time
```

16. Unregistered Active Users

Find users who created an account in the last 30 days but haven't registered for any events.

Query:

```
select user_id, full_name from users  
where registration_date>=curdate()-interval 30 day  
and user_id not in  
(select distinct user_id from registrations );
```

17. Multi-Session Speakers

Identify speakers who are handling more than one session across all events.

Query:

```
select speaker_name, count(session_id) as ns  
from sessions  
group by speaker_name  
having count(session_id)>1;
```

18. Resource Availability Check

List all events that do not have any resources uploaded.

Query:

```
select e.event_id, e.title from events e  
left join resources r  
on e.event_id=r.event_id  
where r.resource_id is null;
```


19. Completed Events with Feedback Summary

For completed events, show total registrations and average feedback rating.

Query:

```
select e.event_id, e.title, count(distinct r.registration_id) as total_registrations,  
avg(f.rating) as avg_rating from events e  
left join registrations r on e.event_id=r.event_id  
left join feedback f on e.event_id=f.event_id  
where e.status='completed'  
group by e.event_id,e.title;
```

20. User Engagement Index

For each user, calculate how many events they attended and how many feedbacks they submitted.

Query:

```
select u.user_id, u.full_name, count(distinct r.event_id) as ea,  
count(distinct f.feedback_id) as fs  
from users u  
left join registrations r on u.user_id=r.user_id  
left join feedback f on u.user_id=f.user_id  
group by u.user_id,u.full_name;
```

21. Top Feedback Providers

List top 5 users who have submitted the most feedback entries.

Query:

```
select u.user_id, u.full_name, count(f.feedback_id) as feedback_count from users u  
join feedback f on u.user_id=f.user_id  
group by u.user_id,u.full_name  
order by feedback_count desc  
limit 5;
```

22. Duplicate Registrations Check

Detect if a user has been registered more than once for the same event.

Query:

```
select user_id, event_id, count(*) as registration_count from registrations  
group by user_id,event_id  
having count(*)>1;
```

23. Registration Trends

Show a month-wise registration count trend over the past 12 months.

Query:

```
select year(registration_date) as yr,  
month(registration_date) as mn,  
count(*) as registration_cnt  
from registrations  
where registration_date>=curdate()-interval 12 month  
group by year(registration_date),month(registration_date)  
order by yr,mn;
```

24. Average Session Duration per Event

Compute the average duration (in minutes) of sessions in each event.

Query:

```
select event_id, avg(timestampdiff(minute,start_time,end_time)) as avg_duration from  
sessions  
group by event_id;
```

25. Events Without Sessions

List all events that currently have no sessions scheduled under them.

Query:

```
select e.event_id, e.title from events e  
left join sessions s on e.event_id=s.event_id  
where s.session_id is null;
```